

C++ Training @ R&S

Klaus Iglberger
27.4.-29.4.2026

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1. Introduction

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Introduction of the Trainer

C++ Trainer/Consultant since 2016

Author of “C++ Software Design”

(Co-)Organizer of the Munich C++ user group

(Co-)Chair of the Software Design at CppCon

Regular presenter at C++ conferences



Klaus Iglberger

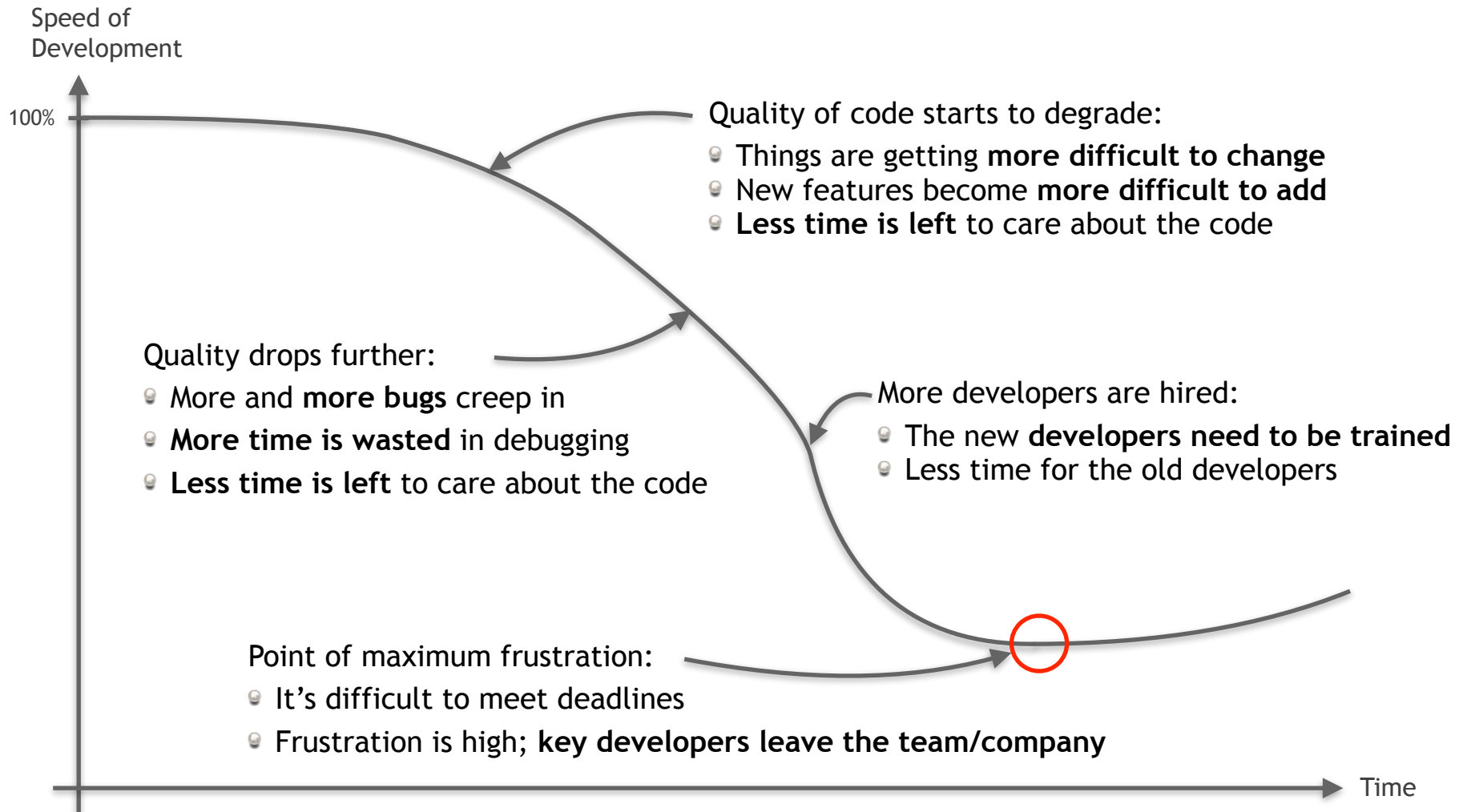
Motivation



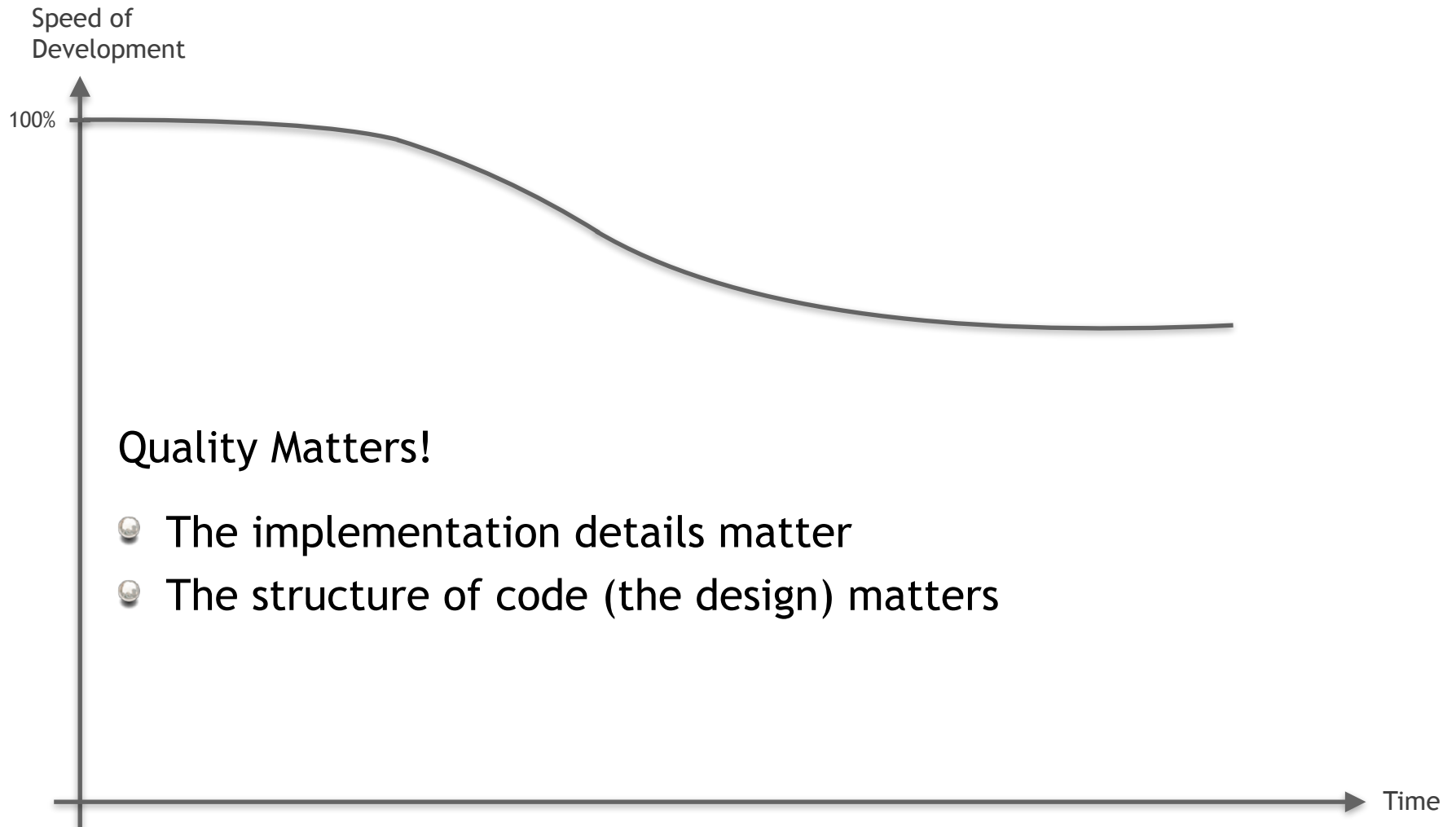
“There is no point in developing software unless you care about doing it well.” (Andrew Hunt, David Thomas, The Pragmatic Programmer)



Motivation



Motivation



Scope of this Training

In this training I will address ...

- ... software design principles and patterns;
- ... design patterns in a multi-paradigm programming language;
- ... classic and modern design patterns;
- ... C++20 Concepts and Ranges;
- ... the philosophy of “Modern C++”.

Scope of this Training

In this training I will NOT address ...

- ... all possible (classic and modern) design pattern;
- ... all variations of design patterns;
- ... all implementation details of modern design patterns;
- ... all C++20 features.

Content of this Training

1. Introduction
2. C++ Software Design
3. C++20 Concepts
4. C++20 Ranges
5. Summary/Literature

Schedule

Monday, 27.4.2026

9:00 - 9:15	<i>Introduction</i>
9:15 - 10:15	<i>C++ Software Design: Motivation</i>
10:15 - 12:30	<i>The Visitor Design Pattern</i>
12:30 - 13:30	<i>Lunch</i>
13:30 - 15:30	<i>The Strategy Design Pattern</i>
15:30 - 17:00	<i>The External Polymorphism Design Pattern</i>

Schedule

Tuesday, 28.4.2026

9:00 - 10:30	<i>The Type Erasure Design Pattern</i>
10:30 - 12:30	<i>Allocation-Free Data Structures</i>
12:30 - 13:30	<i>Lunch</i>
13:30 - 15:30	<i>C++20 Concepts</i>
15:30 - 17:00	<i>The Decorator Design Pattern</i>

Schedule

Wednesday, 29.4.2026

9:00 - 11:00	<i>The CRTP Design Pattern</i>
11:00 - 12:30	<i>The Expression Template Design Pattern</i>
12:30 - 13:30	<i>Lunch</i>
13:30 - 16:45	<i>C++20 Range Algorithms and Ranges</i>
16:45 - 17:00	<i>Summary/Literature</i>

Miscellaneous

I assume you have some experience with C++. If something remains unclear or somewhat vague, **please ask!**

Miscellaneous

Also remember: **You are in control!**

Guidelines

(Core) Guideline: ...

The green boxes represent ...

- ... the most important take-aways;
- ... common idioms and best practice;
- ... markers in the slides.

Programming Tasks

Task (Subchapter/Name): ...

The yellow boxes represent ...

- ... hands-on programming tasks;
- ... links to the provided source code examples.

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